

Questions of the week - XI

Multiple choice Questions

1. Which of the following value of charge is not possible?
 - a. $1.6 \times 10^{-18} \text{C}$
 - b. $3.2 \times 10^{-19} \text{C}$
 - c. $1.6 \times 10^{-20} \text{C}$
 - d. 1.6×10^{-17}
2. If $x = \frac{a \sin \theta + b \cos \theta}{a+b}$ Then
 - a. The dimensions of x and a are same
 - b. The dimensions of a and b are not same
 - c. x is dimensionless
 - d. Dimensions of x and b are same
3. The temperature of a body is measure of the
 - a. Average kinetic energy of the molecules
 - b. Total energy of the molecules
 - c. Potential energy of the molecules
 - d. Velocity of the molecules
4. The dimensions of universal gravitational constant are
 - a. ~~$[M^{-1}L^3T^{-2}]$~~
 - b. $[M^{-1}L^3T^{-1}]$
 - c. $[M^{-1}L^2T^{-2}]$
 - d. $[M^{-2}L^3T^{-2}]$
5. Taking force, length and time as fundamental quantities, dimensional formula for density is
 - a. $[FL^{-4}T^{-2}]$
 - b. $[FL^4T^2]$
 - c. $[FL^{-2}T^4]$
 - d. $[FL^{-4}T^2]$
6. Two concentric spheres are carrying a charge Q and q. When the two spheres are joined by a wire, the charge on the outer surface of the inner sphere is
 - a) Zero
 - b) Q - q
 - c) Q + q
 - d) q - Q

Short answer questions

1. This question is about properties of charge

a. Explain the meaning of the statement "electric charge of a body is quantized."

☒ b. Why can we ignore quantization of charge when dealing with large scale charge?

c. Explain the meaning of Action of point. How this property can be used to our benefit?

d. How can you charge a body positively by induction? Explain.

☒ e. Does charging by friction follow the conservation of charge?

f. Iron chains touching to the ground are suspended from a vehicle carrying inflammable materials, why?

2. This question is about heat and temperature

☒ a. Absolute zero temperature is not zero energy temperature, explain.

b. A Celsius and Fahrenheit thermometers are placed in a water bath. The reading of the Celsius thermometer is one third that of the Fahrenheit thermometer. What is the temperature of the bath? -5.55°C

c. The temperature of body changes by 25°C . What is the corresponding change on the Fahrenheit scale? 45°F

d. Find the temperature which has the same value in Kelvin scale and Fahrenheit scale.

3. This question is about Measurement and dimensional analysis

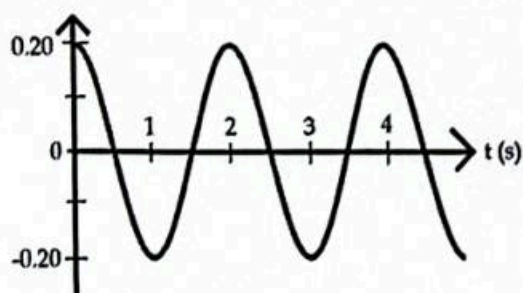
a. A new unit of length is chosen such that speed of light in free space is unity. What is the distance between the sun and the earth in terms of the new unit if light takes 8 min 20 sec to cover this distance? 1.496×10^{11}

☒ b. Find the value of 20J on the system based on 20 cm, 1 Kg and $\frac{1}{2}$ minute as fundamental.

c. A body moving through air at high speed V experiences a retarding force F given by $F = K A \rho V^x$ where A is the surface area of body, ρ is the density of the air and K is numerical constant. Deduce the value of x .

Questions of the week 1 - XII

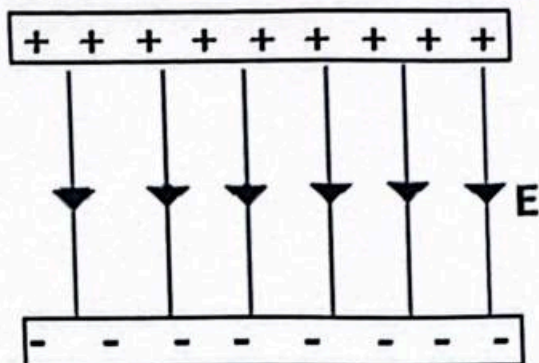
1. a. Define simple harmonic motion (SHM) and explain the variation of velocity with displacement from mean position of a particle in SHM.
- b. The displacement of an oscillating object as a function of time is shown in figure. (Displacement in meter and time in second)



Calculate its

- I) Time period
 - II) Amplitude
 - III) Velocity at mean position and acceleration at extreme.
 - IV) Velocity and acceleration at 10 cm from mean position.
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2. a) Define wave. A wave transmit energy. Does it transmit momentum?
 - b) A displacement of particle is given by: $y = a \sin(\omega t - kx)$. Is it a longitudinal or a transverse wave? Justify.
 - c) A plane progressive wave is represented by the equation: $y = 0.1 \sin\left(200\pi t - \frac{20\pi x}{17}\right)$,
Where y is the displacement in millimeters, t is in seconds & x is distance from a fixed origin O in meters. Find,
 - i) The frequency of the wave, its wavelength & its speed
 - ii) The phase difference between a point 0.25m from Origin and a point 1.10m from Origin.

3. a) Trace the path of an electron and a proton, if both enter a uniform electric field, with the same velocity, perpendicular to the field.



- b) Which particle will have trajectory more curved? Explain.
- c) What is the difference between the deflection of the electron due to electric and magnetic fields?
- d) A beam of electrons, moving with velocity 10^7 m/s, enters midway between two horizontal parallel plates in the direction parallel to the plates which are 5 cm long and 2 cm apart and have a p.d. of V volts between them. Calculate V if the beam is deflected so that it just grazes the edge of the plate. Assume $e/m = 1.76 \times 10^{11}$ C/kg.
4. a) What is Lorentz force? An electron is moving along Y- axis and magnetic field is acting along X-axis. What is the direction of magnetic force acting on it? Explain how Fleming's left hand rule helped to find the direction of force.
- b) A charged particle moves un-deviated through a region in the space. Can you conclude that there is no magnetic field in that region? Explain.